

BOULIA, Australia

WMO: 943330

Lat: 22.912S

Lon: 139.904E

Elev: 158

StdP: 99.44

Time Zone: 10.00 (E10)

Period: 94-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 6.8	(c) 8.0	(d) -10.3	(e) 1.6	(f) 19.4	(g) -8.0	(h) 1.9	(i) 17.9	(j) 8.8	(k) 19.2	(l) 8.0	(m) 19.6	(n) 1.9	(o) 180	(p) 0.394

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 11.7	(c) 41.5	(d) 22.1	(e) 40.4	(f) 22.0	(g) 39.3	(h) 21.9	(i) 26.6	(j) 34.8	(k) 25.7	(l) 33.2	(m) 25.1	(n) 32.5	(o) 2.9	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.9	20.3	29.5	24.1	19.3	28.5	23.2	18.4	28.6	84.4	35.0	80.4	33.0	77.7	32.3	33.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.1	(b) 8.6	(c) 7.7	(e) 4.1	(f) 43.3	(g) 1.1	(h) 0.9	(i) 3.3	(j) 43.9	(k) 2.7	(l) 44.4	(m) 2.1	(n) 44.9	(o) 1.3	(p) 45.6		
(4) 10.1	(5) 8.6	(6) 7.7	(7) 4.1	(8) 43.3	(9) 1.1	(10) 0.9	(11) 3.3	(12) 43.9	(13) 2.7	(14) 44.4	(15) 2.1	(16) 44.9	(17) 1.3	(18) 45.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.9	31.6	31.5	29.4	25.2	20.0	17.1	16.5	18.4	23.4	26.2	28.9	30.7		
	DBStd	6.33	2.94	2.46	2.63	3.42	3.11	3.49	3.15	3.37	3.84	3.47	3.34	3.00		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	208	0	0	0	2	19	65	76	41	6	1	0	0		
	CDD10.0	5426	668	602	600	456	311	213	202	261	402	503	566	642		
	CDD18.3	2594	410	368	342	208	71	29	20	43	158	245	316	383		
	CDH23.3	35963	6321	5560	4741	2340	619	254	209	548	1947	3212	4474	5738		
	CDH26.7	20892	4082	3496	2717	1133	158	53	33	174	962	1769	2689	3626		
(14) Wind	WSAvg	2.8	2.6	2.7	2.7	2.3	2.2	2.1	2.1	2.8	3.2	3.6	3.4	3.3		
(15) Precipitation	PrecAvg	259	53	49	28	14	14	8	10	6	6	15	23	36		
	PrecMax	774	465	229	254	117	106	60	116	38	40	83	151	259		
	PrecMin	51	0	0	0	0	0	0	0	0	0	0	0	0		
	PrecStd	167	75	60	43	27	24	14	21	10	10	20	34	45		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.7	41.7	40.0	37.4	31.6	30.1	29.5	32.9	37.8	40.4	41.7	42.9		
		MCWB	22.4	22.8	20.8	21.3	17.2	18.5	15.5	17.9	19.2	21.2	21.3	21.2		
	2%	DB	41.2	40.3	38.4	35.5	30.2	28.3	27.1	30.7	35.4	38.4	40.4	41.4		
		MCWB	22.6	22.9	21.2	20.1	17.4	17.7	15.2	17.0	19.0	20.5	21.4	21.9		
	5%	DB	39.9	39.1	37.3	34.0	28.9	26.6	25.6	28.9	34.1	36.6	38.9	40.0		
		MCWB	22.1	22.7	21.1	19.9	17.1	16.5	14.5	15.9	18.5	19.5	21.0	22.1		
	10%	DB	38.6	37.8	35.9	32.6	27.2	24.3	23.9	26.8	32.4	35.0	37.1	38.5		
		MCWB	22.0	22.5	20.8	18.8	16.4	15.1	13.4	14.6	17.5	18.5	20.6	21.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.4	28.0	26.1	25.5	21.0	20.1	18.8	20.5	23.2	25.5	27.6	27.2		
		MCDB	37.0	36.7	30.9	32.7	25.1	28.0	25.4	28.5	34.4	37.8	35.5	38.6		
	2%	WB	26.3	26.4	24.9	23.1	19.4	18.6	16.9	18.7	21.6	23.1	24.6	25.7		
		MCDB	34.4	33.4	30.4	29.5	26.2	26.6	24.9	28.0	31.4	32.8	33.8	34.5		
	5%	WB	25.3	25.6	23.9	21.9	18.4	17.3	15.4	17.0	20.2	21.6	23.5	24.9		
		MCDB	33.2	32.5	30.7	30.0	25.9	24.9	24.2	26.4	30.6	31.0	31.9	32.4		
	10%	WB	24.6	25.1	23.0	20.8	17.4	16.1	14.2	15.6	18.8	20.3	22.6	24.2		
		MCDB	32.8	31.9	31.7	29.7	25.2	23.2	22.8	25.5	29.3	30.5	31.2	31.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.7	11.5	11.6	11.8	11.7	12.1	13.1	13.9	14.3	13.8	13.1	12.3		
		MCDBR	13.3	12.8	13.1	13.5	13.0	13.6	14.7	15.7	15.9	15.6	15.6	15.2		
	5% WB	MCWBR	4.8	5.2	5.3	5.9	6.1	6.6	7.7	8.0	7.4	7.0	6.9	6.6		
		MCWBR	11.1	10.3	9.6	11.2	10.4	12.1	12.6	13.9	13.9	12.4	12.4	11.0		
(40) Clear-Sky Solar Irradiance	taub		0.383	0.376	0.357	0.340	0.326	0.310	0.309	0.324	0.369	0.374	0.388	0.393		
	taud		2.436	2.454	2.478	2.480	2.470	2.497	2.472	2.406	2.306	2.314	2.331	2.362		
	Ebn at Noon		961	955	945	920	892	888	902	922	918	945	949	953		
	Edh at Noon		123	119	111	102	94	88	93	108	129	135	136	133		
(44) All-Sky Solar Radiation	RadAvg		7.47	7.39	6.67	5.85	4.77	4.25	4.61	5.62	6.57	7.43	7.65	7.57		
	RadStd		0.54	0.45	0.44	0.42	0.26	0.25	0.27	0.23	0.46	0.50	0.38	0.56		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page